

Math 213 – Elementary Linear Algebra

Section 5.4: Orthogonal Diagonalization

From orthogonality to diagonalization

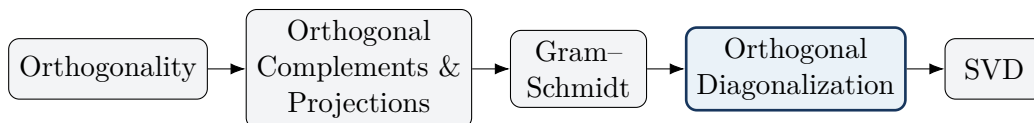
Last Time

Last time we learned how to use the Gram–Schmidt Process to construct orthogonal and orthonormal bases.

Today

Today we combine those ideas with diagonalization.
By the end of today's lesson you should be able to

- Orthogonally diagonalize a symmetric matrix.
- Use the Gram–Schmidt Process to orthogonalize repeated eigenvectors.
- Construct a symmetric matrix with prescribed eigenvalues and orthonormal eigenvectors.
- Use the Spectral Theorem to prove a matrix is orthogonally diagonalizable.



Eigenvalues and Eigenvectors of Symmetric Matrices

Recall: Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Theorem

Let A be a **symmetric** matrix.

1. If the entries of A are **real** (not imaginary / complex), then the eigenvalues of A are also **real** (not imaginary / complex).
2. Any two eigenvectors of A corresponding to **distinct** eigenvalues are **orthogonal**.

Example

Find an eigenvector of $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$ for each eigenvalue ($\lambda = -3, 2$). Are they orthogonal?

Orthogonal Diagonalization

Definition

A square matrix A is called **orthogonally diagonalizable** if there exists an *orthogonal* matrix Q and diagonal matrix D such that

$$Q^T A Q = D.$$

Since $Q^{-1} = Q^T$, this is simply a special kind of diagonalization.

Example

Orthogonally diagonalize $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$, from above.

Strategy

If a symmetric matrix has distinct eigenvalues, the eigenvectors are already orthogonal, so you only need to normalize each eigenvector and place them as the columns of Q and order D correctly.

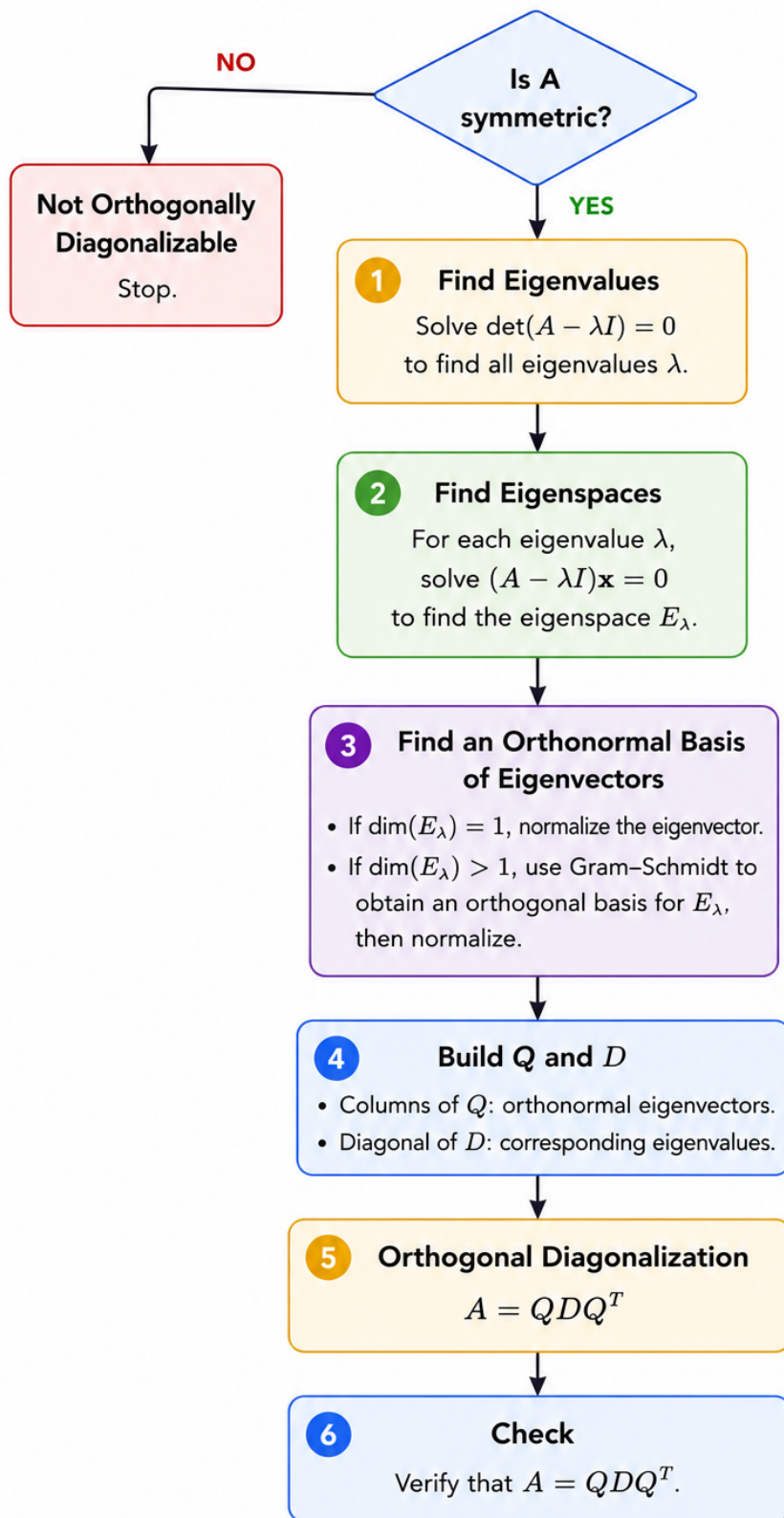
Theorem (Spectral Theorem)

A matrix is orthogonally diagonalizable if and only if it is symmetric.

Example

Prove that if A and B are orthogonally diagonalizable, so is $A + B$.

ROADMAP: Orthogonal Diagonalization of a Symmetric Matrix



Example

Orthogonally diagonalize A given that $p_A(\lambda) = (\lambda + 1)^2(\lambda - 5)$.

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Constructing a Matrix with Given Eigenvalues and Eigenvectors

Strategy

To construct a symmetric matrix with specific eigenvalues $\lambda_1, \dots, \lambda_k$ and orthogonal eigenvectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$:

1. Normalize the eigenvectors for $i = 1, \dots, k$

$$\mathbf{q}_i = \frac{1}{\|\mathbf{v}_i\|} \mathbf{v}_i$$

2. Form $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_k]$.
3. Form $D = \text{diag}(\lambda_1, \dots, \lambda_k)$.
4. Compute

$$A = QDQ^T.$$

Example

Construct a symmetric 2×2 matrix with eigenvalues $\lambda_1 = 4$, $\lambda_2 = 0$, and corresponding orthogonal eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Spectral Decomposition

Theorem (Spectral Decomposition Theorem)

Let A be a symmetric matrix with **orthonormal** eigenvectors $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ and corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Then

$$A = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^T + \lambda_2 \mathbf{q}_2 \mathbf{q}_2^T + \cdots + \lambda_n \mathbf{q}_n \mathbf{q}_n^T.$$

We call this expansion the **spectral decomposition** of A .

Example

Find a spectral decomposition of $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$.

Reflection

In your own words, what makes orthogonal diagonalization different from ordinary diagonalization?

Section 5.4 Summary

Today we learned how to:

- Orthogonally diagonalize a symmetric matrix.
- Use Gram–Schmidt when an eigenspace has dimension greater than one.
- Construct a symmetric matrix from its eigenvalues and eigenvectors.
- Write the spectral decomposition of a matrix.

Next Time

Next we move to the Singular Value Decomposition (SVD), which extends many of the ideas of orthogonal diagonalization to general matrices.